

JOSHUA LIU

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EDUCATION

University of Illinois Urbana-Champaign

Urbana, IL

Bachelor of Science in Computer Science

Expected May 2030

- Relevant coursework: Intro to Computer Science I & II, Discrete Structures, Calculus III, Linear Algebra

Lexington High School

Lexington, MA

High School Diploma | GPA: 3.92

August 2022 - May 2026

EXPERIENCES

Illini Electric Motorsports

University of Illinois Urbana-Champaign

Software Team Member

August 2026 - Present

- Developing the protocols and firmware for a Hardware In the Loop (HIL) testing board using C, enabling strict testing of safety-critical PCBs and firmware without connecting them to the physical vehicle

LexVex Robotics

Lexington High School

Captain, Programmer

October 2022 - May 2026

- Captain of 5 Lexington High School VEX Robotics teams (15797A, 15797H, 15797X, 15797Z, and 66659A)
- Researched, architected, and implemented complex localization and motion algorithms such as Monte Carlo Localization and RAMSETE to reach near-perfect levels of autonomous consistency
- Pioneered the use of Rust in VEX Robotics to prevent fatal memory errors and increase algorithm performance
- 2025-2026 Massachusetts State Champions, won ten other tournament championships

Boston College

Newton, MA

Computer Science Department Intern

May 2025 - August 2025

- Researched and studied economics and simulations to find ways of reducing economic exploitation
- Developed an agent-based model using C++ to evaluate the feasibility of alternative economic distribution systems

LIT Programming Competition

Lexington, MA

Organizer, Problem Writer

September 2022 - August 2026

- Organized and ran a yearly programming competition for thousands of students across 6 continents worldwide
- Created 30+ Capture the Flag and competitive programming problems for the competition

PROJECTS

MochBot

- Developed and deployed a versus Tetris bot for high-level competitive training using a BFS move generator paired with beam search and neural network evaluation mechanism that consistently beats top human players
- Architected a distributed TypeScript network to connect 30+ simultaneous C++ engine instances to game servers, deploying across multiple nodes to support 1,400+ weekly active users and 40,000+ games/week

Zoron.app

- Developed user-friendly cross-platform replacement for the Follet Software's Aspen Student Information System
- Implemented Aspen-compatible authentication and client-side data retrieval, allowing students to securely access schedules, grades, transcripts, and attendance from a unified dashboard
- Utilized SvelteKit, Typescript, and MongoDB to build a responsive full-stack application used by hundreds of students at Lexington High School on a daily basis

Triangle.js

- Engineered an open-source package for interacting with the private TETR.IO game API
- Reverse-engineered the API to develop a fully compatible headless Typescript client, reaching 30+ stars on Github and 150+ weekly downloads on NPM, and being authorized for use by the game's creators

TECHNICAL SKILLS

- **Languages:** Rust, C/C++, Python, TypeScript, JavaScript, HTML/CSS, Java
- **Frameworks & Libraries:** SvelteKit, Next.js, React, Node.js
- **Developer Tools & Databases:** Git, GitHub, MongoDB, Supabase, S3, Docker, Linux, REST APIs